

1 LEMMA 4.2, FORMALIZED

1.1 The aggregate factoring as a coordinate-by-coordinate identity of linear maps

The Sofa Theorem — formal development of Lemma 4.2 (the collar’s aggregate factoring) · 9 July 2026 Purpose: replace the prose of the announcement’s Lemma 4.2 with explicit formulas, so that the multi-heavy concern is settled by inspection. Nothing here changes the statement; it changes the level of detail.

1.1.1 0. Data and conventions

$q = 3^v - 9$, $f \in \{0, 1, 2, 3\}$. $\Delta = (\dots) (S_{\{q+f\}})$ with coefficients $\underline{c}_i = \sum_m c_{\{i, m\}} s^m$ (m an exponent vector, $|m| = q+f$). Sheets $J = \{P, P, P\}$ with ordered pairs $P_k = (a_k, b_k)$, $a_k < b_k$; sheet coordinates u_k via $s_{\{a_k\}} = u_k$, $s_{\{b_k\}} = -u_k$.

Restriction formula (monomial-wise, exact): for any exponent vector m , $\Delta|_J = (-1)^{\sum_k m_{\{b_k\}}} \cdot \prod_k u_k^{m_{\{a_k\}} + m_{\{b_k\}}}$. (R)

Extraction: $\underline{c}_{\{k, l\}}^{\{J\}} :=$ the coefficient of u_1^q in $\Delta_k = (\underline{c}_{\{a_k\}} - \underline{c}_{\{b_k\}})|_J$, i.e. the sum of all terms of Δ_k with u_1 -exponent q , divided by u_1^q . (Well-defined: by $\deg = q+f < 2q$ at most one coordinate can carry exponent q per monomial.)

1.1.2 1. Which monomials contribute, and through what functional

Fix J , fix l , and let $p = P_1 = \{a, b\}$ ($a < b$). By (R), a monomial s^m of $\underline{c}_i$ contributes to the u_1^q -part of $\underline{c}_i|_J$ iff its **p-pair-weight** $w_p(m) := m_a + m_b = q$. Write such an m as $(m_a, m_b, \dots)$ with $\dots$ = the restriction of m to the four variables outside p , and $w := m_a + m_b = q + t$. Since $|m| = q+f > t = f - |l|$ is **DETERMINED** by $\dots$. (T) By (R), the contribution of s^m to the u_1^q -part of $\underline{c}_i|_J$ is $> (-1)^{m_b} \cdot (-1)^{\{(\dots, J)\}} \cdot u_1^t \cdot (\dots, J)$, where $(\dots, J) = \underline{c}_{\{k, l\}} u_k^{\{a_k\} + \{b_k\}}$ and $(\dots, J) = \sum_{x \in \text{supp } \dots} x = \text{some } b_k - \dots_x$ — both depending only on $(\dots, J)$, not on the split (m_a, m_b) .

Hence the split-dependence enters ONLY through $(-1)^{m_b}$. Define the alternating aggregates (linear functionals on the coefficients of $\underline{c}_i$): $> \check{c}_i(p, \dots) := \sum_{\{ \dots = 0 \}} (-1)^{\dots} \cdot c_{\{i, (w-\dots, \dots)\}}$, with $w = q + f - |l|$. (AGG) Then, coordinate-by-coordinate: $> [u_1^q\text{-part of } \underline{c}_i|_J] = \sum_{\dots} (-1)^{\{(\dots, J)\}} \cdot \check{c}_i(p, \dots) \cdot u_1^{\{f-|l|\}} \cdot (\dots, J)$. ()

1.1.3 2. The factoring, as an identity of maps

Let $W_f := \underline{c}_{\{i, p, \dots\}} F$ (i 6 slots, p 15 pairs, $\dots$ a monomial of degree f on the 4 complement variables) — the FREE aggregate space, q -free. Define: $- \text{agg}: (S_{\{q+f\}}) \rightarrow W_f$, $\check{c}_i(p, \dots) \mapsto \underline{c}_{\{i, p, \dots\}}$ — a tuple of linear functionals (AGG); $- A_f: W_f \rightarrow \underline{c}_{\{J, k, l\}} F [u]_f$ by $A_f(\underline{c}_{\{i, p, \dots\}}) = \sum_{\{J \dots p\}} \sum_{\{k, l(J, p)\}} \underline{c}_i^{\{J, k\}} (-1)^{\{(\dots, J)\}} u_1^{\{f-|l|\}} \cdot (\dots, J)$ in coordinate (J, k, l) , where $l = l(J, p)$ is p ’s position in J and $\underline{c}_i^{\{J, k\}} = +1$ if $i = a_k$, -1 if

$i = b_k, 0$ otherwise. Both maps have **fixed, q-free coefficients** (± 1 and monomial placements determined by $(, J, i)$ alone).

Proposition (factoring). $\Phi_{\text{collar}} = A_f \text{ agg}$ as linear maps on $(S_{\{q+f\}})$. Consequently $\text{Im}(\Phi_{\text{collar}}) \subseteq \text{Im}(A_f)$; combined with T-block (iii) (antisymmetry, forced for syzygies), $\text{Im}(\Phi_{\text{collar}}|_{\{\text{Syz}\}}) \subseteq D_f := \text{Im}(A_f) \cap \{\text{antisym}\}$. *Proof.* Evaluate both sides on $_$ at each coordinate (J,k,l) : the left side is $_k,l^{\{(J)\}} = [u_l^{\text{q-part of } (_a_k - _b_k)}]_J$; by $(_)$ applied to $i = a_k$ (sign $+1$) and $i = b_k$ (sign -1), this equals the (J,k,l) -coordinate of $A_f(\text{agg}(_))$. The two sides agree on every $_$ and every coordinate.

1.1.4 3. Three remarks that settle the standing objections

Remark 1 (multi-heavy monomials — the objection, dissolved). A monomial may satisfy $w_p(m) \leq q$ for several pairs p (e.g. one exponent $\leq q-f$). This creates NO ambiguity: **agg** is a tuple of functionals — the SAME coefficient $c_{\{i,m\}}$ simply enters the aggregates of every pair for which m is heavy, each time with the sign dictated by (AGG). No disjoint decomposition of $_$ is used anywhere; the factoring identity is verified coefficient-wise in the proof above. (A sign inconsistency would require the same coefficient to enter one coordinate of the output through two different functionals with contradictory signs — impossible, since each output coordinate (J,k,l) reads only the $p = P_l$ aggregates, once.) **Remark 2 (why the bound needs nothing more).** The ceiling uses ONLY $\dim \text{Im}(\Phi|_{\text{Syz}}) \leq \dim D_f$ — a superset bound. No injectivity, surjectivity, or exactness of **agg** or A_f is claimed or needed. $\dim D_f = 10, 55, 145, 280$ is a fixed rank certificate, verified four independent ways (including Macaulay2). **Remark 3 (the kernel identification uses radicality, NOT Recognition out of range).** $\ker(\Phi_{\text{collar}}|_{\{\text{Syz}\}})$: for a syzygy, T-block (i) kills the reduced parts and $_ = 0$ kills the large parts, so $\Delta_k = 0$ identically on every sheet, for all k, J . Then $_a - _b$ vanishes on all three sheets through each edge p , i.e. on $V(p)$; since $I(p)$ is RADICAL (mod $_$, $_ = -(c+d)(c+e)(d+e)$ splits into three distinct linear forms in char 3), $_a - _b \in I(V(p)) = I(p)$, i.e. $_ \in M$. The Recognition Theorem (proved for $e < q$) is never invoked at $e = q$. Conversely $M \subseteq \ker$ trivially. Hence $\ker = (M)$, of dimension $\text{HF}(R)_{\{q+f\}} + __{\{q+f\}}$ by the certified census law (valid e).

Part of the Sofa Theorem repository: github.com/tretoef-estrella · tretoef@gmail.com